

Name: Dog Paw Washer _____ Date: 3/13/2026 _____
_____pd 2_____

Class:

MyDesign Portfolio

Element D Initial Template

Element D: Design Concept Generation, Analysis, and Selection

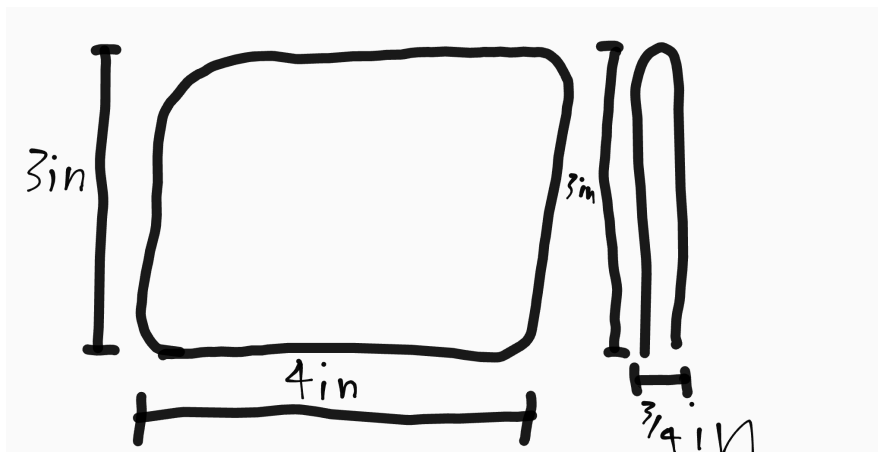
Design Idea Generation (D1)

Design Idea #1

Date: 3/13/2026

Description: An abrasive towel with natural materials that can clean a dog's paws.

Drawing that includes distinguishing information such as scale, dimensions, and materials:



Design Idea #2

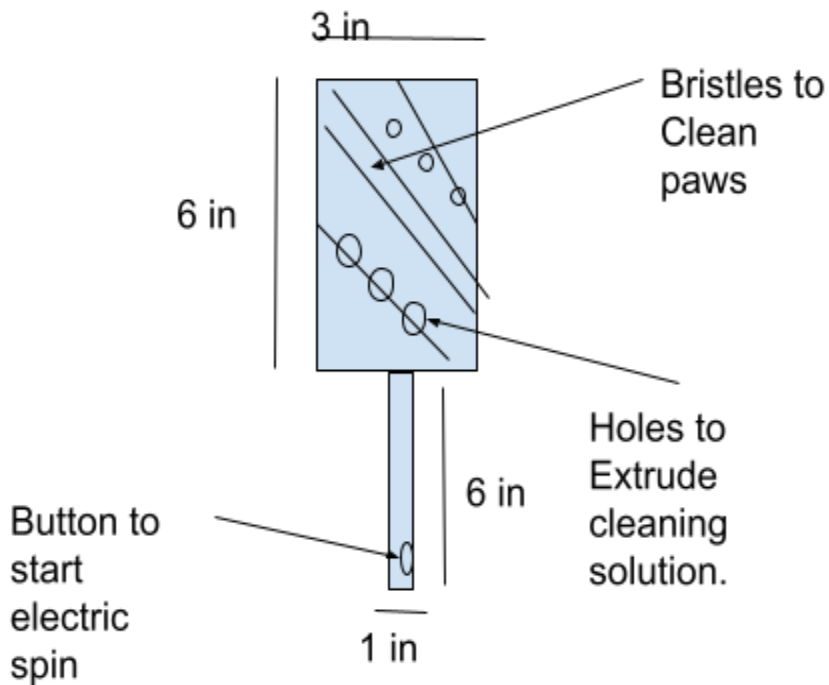
Date: 3/13/2026

Description: An electric brush that spins at one end with an abrasive towel that can clean a dog's paws.

Drawing that includes distinguishing information such as scale, dimensions, and materials:

Name: Dog Paw Washer _____ **Date:** 3/13/2026 _____
_____pd 2_____

Class:



Design Idea #3

Date: 3/13/2026

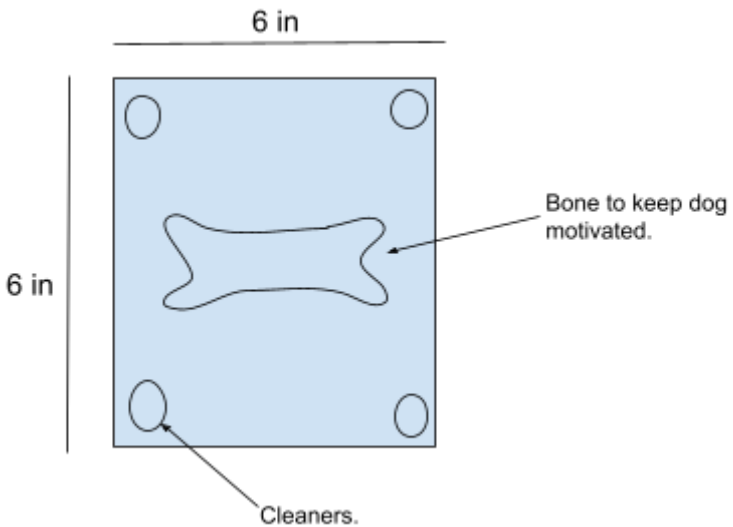
Description: A machine that attracts dogs to stay where they are, stand in a basin, that can mechanically clean and dry their paws.

Drawing that includes distinguishing information such as scale, dimensions, and materials:



Name: Dog Paw Washer _____ **Date:** 3/13/2026 _____
_____pd 2_____

Class:



Design Idea #4

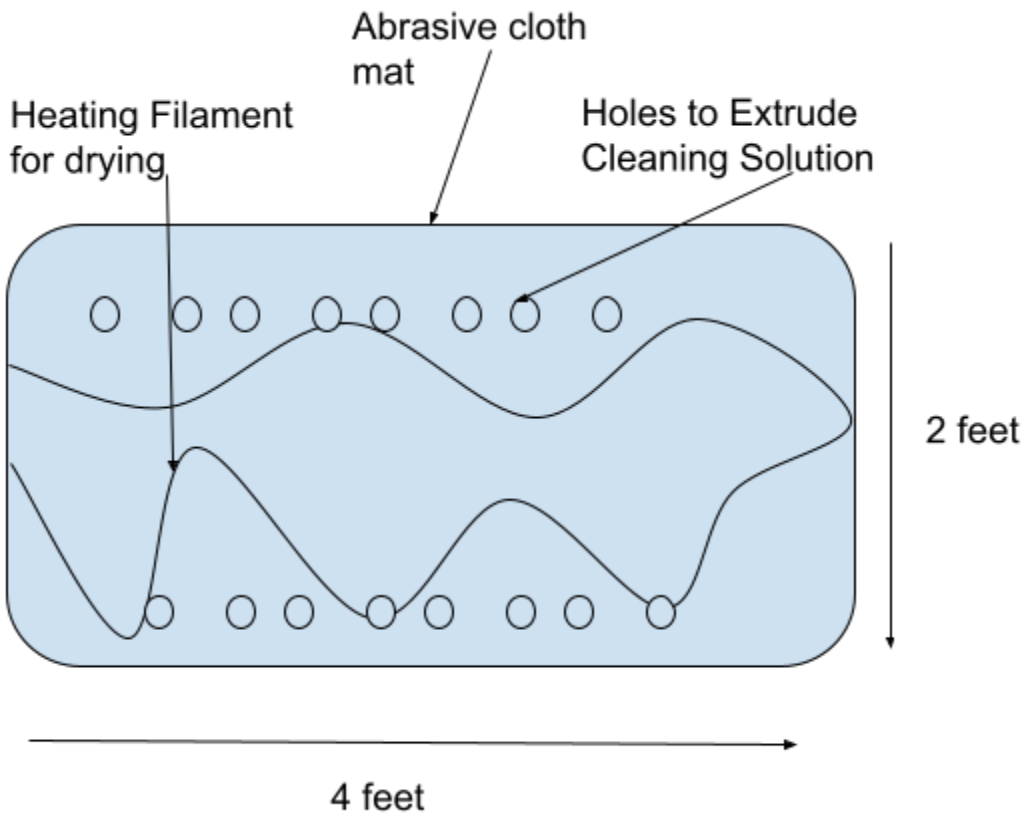
Date: 3/13/2026

Description: A dog mat that is dry on one side touching the ground, and wet on the other side facing up. The mat has the ability to excrete washing solutions to clean the dog's paws as much as possible. The mat has a built-in heater that dries the paws.

Drawing that includes distinguishing information such as scale, dimensions, and materials:

Name: Dog Paw Washer _____ **Date:** 3/13/2026 _____
_____pd 2_____

Class:



Design Idea Comparison and Selection (D1, D2)

	weight	Option 1	Option 2	Option 3	Option 4
Criterion A	10	-	+	+	-
Criterion B	9	+	-	+	+
Criterion C	8	-	-	+	+
Criterion D	7	+	+	+	+

Name: Dog Paw Washer _____ **Date:** 3/13/2026 _____ **Class:**
____pd 2_____

Criterion E	6	-	+	+	-
score		16	23	40	24

Our team chooses to design a machine that attracts dogs to stay where they are, stand in a basin, that can mechanically clean and dry their paws to solve the community problem.

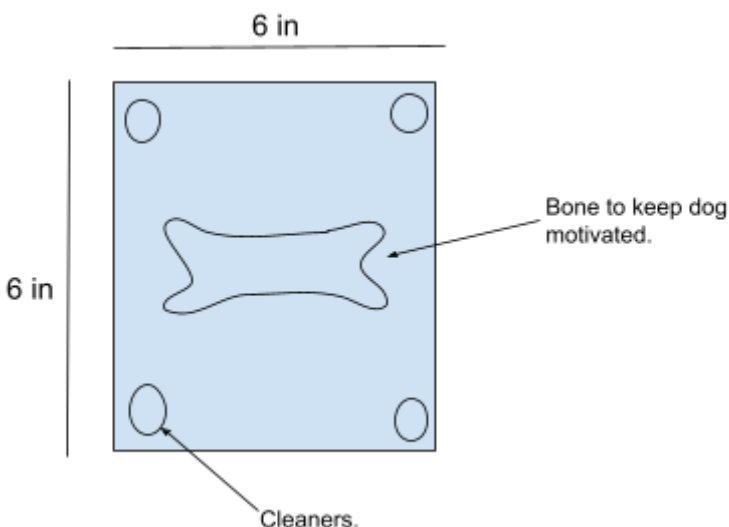
We choose to build a machine that attracts dogs to stay where they are, stand in a basin, that can mechanically clean and dry their paws, because this design idea realistically meets all of the high and low priority requirements outlined for our project, more so than other design ideas.

Design Blueprint (D4)

Design Blueprint #1

Date: 3/13/2026

Sketch:



Manufacturing Methods:

Step 1: Use cardboard cutouts to prototype main pieces of assembly

Step 2: Find plastic or vinyl pieces to create waterproofing

Name: Dog Paw Washer _____ **Date:** 3/13/2026 _____ **Class:**
____pd 2_____

Step 3: Cut out specific openings to create first iteration functionality
Supply List and Potential Costs:

Item	Cost each	# needed	Total
Cardboard	\$0 (Provided by school)	Probably 10's of sq ft.	
Tape	\$0 (Provided by school)	Unknown	
String	\$0 (Provided by school)	Unknown	
Scissors	\$0 (Provided by school)	Just 1 pair	
Total Cost	\$0 (Provided by school)		

Plan of Action (D3)

We will use this Gantt Chart to Keep track of our Plan of Action:

<https://docs.google.com/spreadsheets/d/1bUxz7fIUHUs0P6FUe1HOuxBnz-LqLuOSyJZI0A8juBQ/edit?gid=1115838130#gid=1115838130>

Iterations for the Solution:

Design Blueprint #2

Date:

Sketch:

Manufacturing Methods:

Step 1:

Step 2:



Name: Dog Paw Washer _____ **Date:** 3/13/2026 _____ **Class:**
____pd 2_____

Step 3:
Supply List and Potential Costs:

Item	Cost each	# needed	Total
Total Cost			

Supply List and Potential Costs:

Item	Cost each	# needed	Total
Total Cost			